

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (C.E.)

April/May 2012

CE-302 System Programming (S.P.)

Date: 02.05.2012, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Rough work is to be done in the last page of main supplementary, please don't write anything on the question paper.
5. Indicate clearly, the option(s) you attempt along with its respective question no.
6. Figures to the right indicate marks.

SECTION-I

Q-1 Answer the following questions.

1. What do you mean by a translator? Give example of various translators with their input and output. 3
2. What do you mean by macro? What is macro expansion? 4
3. What are the advantages of high level and assembly level languages over machine language? 4

Q-2

- [A] Write elements of assembly language with example. 4
- [B] Define process and thread. Explain multi-processing and multi-threading in brief. 4
- [C] What is back patching? Why is it needed? Explain in brief. 4

OR

Q-2

- [A] Explain all the data structures required in single pass assembler. 4
- [B] Differentiate between system software and middle ware. 4
- [C] Differentiate between instruction set and addressing modes. 4

Q-3

- [A] What is the relationship between application software, system software, operating system, utility programs and the machine structure? Give an example for the same. 4
- [B] Explain the differences between relocatable and non-relocatable program. 4
- [C] Define lexical and semantic expansion. Write a macro which takes A, B, C and D as parameters and calculates $A*B+C*D$ in AREG. 4

OR

Q-3

- [A] Write down algorithm for program relocation. 4
- [B] What is the content of symbol table? What is your choice as a data structures to implement the symbol table? Justify your answer. 4
- [C] What is macro preprocessor? How does it work? 4

SECTION-II

Q-4

1. Explain the task of following data structures. 3
 - a) Language Translator
 - b) Linker
 - c) Loader
2. Explain all data structures required in multi pass assembler? 4
3. What do you mean by debugging a program? What are the methods available for debugging? 4

Q-5

- [A] What do you mean by program relocation? What is known as linking loader? 4
- [B] What is the difference between scanning and parsing? What is the communication channel between scanner and parser? What is the role of symbol table in it? 4
- [C] What is Editor? What are the functions of an Editor? 4

OR

Q-5

- [A] What is the role of a preprocessor in compiler? What are the major functions done by the Lexical Analyzer? 4
- [B] Explain ENTRY and EXTERN statement with example. Take two assembly language modules M1 and M2. M1 should contain ENTRY statement and M2 should contain EXTERN statement. 4
- [C] What is binary program? Elaborate in brief. 4

Q-6

- [A] Explain how the nested macro call is processed? How stack is used to manage the data structures of macros in nested call. 4
- [B] What is the function of loader? Explain Dynamic Loading in brief. 4

OR

- [B] Write down algorithm for program relocation. 4
- [C] Explain following terms with their significance: 4
 - a) Relocation
 - b) Address sensitive instruction
 - c) Address sensitive Program
 - d) Translated Origin

OR

- [C] Explain ENTRY and EXTERN statement with example. Take two assembly language modules M1 and M2. M1 should contain ENTRY statement and M2 should contain EXTERN statement. 4